

Aidan O'Brien

Madison, Wisconsin / Oceanport, New Jersey

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EDUCATION

University of Wisconsin-Madison

Bachelor of Science in Computer Science

GPA: 3.97, Dean's List

Madison, WI

Graduating May 2027

Relevant Coursework: Algorithms, Operating Systems, Big Data Systems, Artificial Intelligence, Computer Organization, Intro to Computer Engineering, Discrete Math, Programming 2 & 3, Linear Algebra, Calc 2 & 3

EXPERIENCE

Software Engineer Intern

May 2025 – Present

CargoLabs

Remote

- Eliminated **8+ hours/month** of manual work by automating bordereau reporting with an **AWS + SQL cron** job.
- Standardized authentication pipelines across **7 AWS Cognito** user pools while implementing consistent UI design, creating unified user workflows and reducing authentication complexity across all portals.
- Redesigned **25+** legacy Bootstrap UI components with **Shadcn/UI + TailwindCSS** while relocating components outside **Next.js** monorepo architecture, creating modern UX with better reusability and developer experience.
- Built comprehensive customer service portal enabling the CargoLabs team to efficiently manage and troubleshoot client website problems, reducing average issue resolution time by **45-60%**.

Software Engineer Intern

June 2024 – Aug. 2024

Collectwise

Remote

- Developed full-stack for Collectwise, a **Y Combinator**-backed debt collection **startup** powered by generative AI.
- Automated debtor reports by analyzing 10,000+ **Firestore** documents with **Node** endpoints.
- Created the file upload and download system seen on the website with **React** and **Firebase**.

PROJECTS

BadgerBase 🐾 | *Next.js, Node.js, MySQL, Vercel, GitHub Actions*

July 2025 – Present

- Created a web app that has reached **1800+ active users** in just **4 days** after launch that optimizes the UW Madison course sign-up experience by integrating course, professor, and grade data from multiple sources.
- Reverse engineered UW APIs to build daily data collection pipelines using **GitHub Actions** for reliable updates.
- Designed backend with **MySQL** on **Railway** to serve enrollment and grade data for over **20,000** courses
- Built scalable API endpoints and responsive **Next.js** frontend with Shadcn able to handle peak enrollment usage.

Flask Investment Calculator 🐾 | *Python, SQLite, HTML/CSS, Pandas/Matplotlib, Jinja*

Jan. 2024 - March 2024

- Set up a financial forecasting tool that visualizes investment growth with interactive **Matplotlib** graphs and stylized year-over-year data tables.
- Stored and retrieved user data with **SQLite** while transferring information from **Flask** backend to HTML frontend.

LEADERSHIP

Cast Director

June 2024 – May 2025

Humorology

Madison, WI

- Lead a cast of students that raised **\$75,000** for a local charity in an annual theatrical charity competition.
- Co-wrote the script for the team's theatrical production, showcasing creativity and ingenuity.
- Honed **leadership** and **project management** skills by consistently driving timely completion of team objectives.

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, C, SQL, TypeScript, HTML/CSS, Bash, Assembly

Dev Tools: AWS, Docker, GitHub Actions, MySQL, Linux, Firebase, Git, Postman, Vercel, Cursor, Railway

Frameworks: Next.js, React, Flask, Node.js, Tailwind, Bootstrap, Express, Hono

Machine Learning : Sklearn, Pytorch, Keras, Matplotlib, HTC, Seaborn, Pandas, Numpy